Multiscale Knowledge Distillation-Aware Classification for Remote Sensing Images

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***Abstract*—Knowledge distillation is a technique designed to enhance the recognition capabilities of lightweight neural networks, which are suitable for deployment on mobile devices. This approach effectively addresses the practical demands of the Internet-of-Things (IoT) era. The inherent complexity of remote sensing data representations presents a significant challenge to the effective distillation from complex teacher networks to lightweight student networks. To address this issue, a novel multiscale contrastive learning knowledge distillation model has been introduced and successfully implemented in remote sensing data classification. This novel distillation model effectively transfers the recognition capabilities of complex deep neural networks to lightweight networks by introducing a multiscale contrastive learning supervision module (MCSM) and a transformation consistency supervision module (TCSM). The MCSM primarily provides multiscale cognitive guidance to the student network. This module enhances the student network's overall understanding of intraclass similarities and interclass differences in complex scenes through contrastive learning. The TCSM mainly provides cognitive guidance on target size changes for student networks. This module enhances the transfer ability of the teacher network to the student network on varying target sizes. The experimental results on standard datasets, including UC Merced, AID, OPTIMAL-31, and NWPU-RESISC45, confirm the effectiveness of the proposed model. The experiment demonstrates that the proposed method achieved a classification accuracy that is 1% higher than that of existing distillation models. Furthermore, the classification accuracy of ResNet34 utilizing this distillation method is 0.4%–3.1% higher than that of the improved ResNet34 network, which has a greater number of parameters.**

***Index Terms*—Contrastive learning, knowledge distillation, multiscale features, remote sensing images classification.**

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I. Introduction

THE effective interpretation and analysis of remote sensing data are urgent tasks in many research fields, including agriculture [1], military applications [2], geological exploration [3], [4], and disaster warning and prediction [5]. As a fundamental task of remote sensing data interpretation, remote sensing data classification has garnered significant attention. Recognizing the significant social impact of remote sensing data interpretation, scholars are currently focused on analyzing and researching this area using deep neural network models [6], [7], [8], [9]. Subsequently, the cognitive abilities of deep neural networks in interpreting remote sensing data are continuously improving, as evidenced by recent studies [10], [11], [12], [13]. In theory, the strong approximation and fitting capabilities of deep neural networks are the fundamental attributes that enable them to achieve optimal classification results in complex remote sensing data scenarios [14], [15], [16]. However, in the era of the Internet of Things (IoT), it is essential to transfer high-performance deep neural networks from laboratory servers to mobile terminal devices. Consequently, achieving high-performance interpretation of lightweight networks that can be deployed on edge devices presents a significant challenge.

To achieve these tasks, enhancing the interpretability of lightweight networks has become a key area of research [17]. There are several model compression methods to enhance the performance of lightweight neural networks, including model quantification, low-rank decomposition, network pruning, and neural network structure search. In contrast to these methods, knowledge distillation offers the flexibility to choose the structural framework of the lightweight network. Therefore, models that utilize knowledge distillation strategies to improve interpretative performance on remote sensing data have been proposed [18], [19], [20], [21]. However, VSDNet [18] enhances the network's generalization ability through self-distillation. The ET-GSNet model [20] combines the recognition capabilities of Transformers and convolutional neural networks (CNNs) using distillation strategies. Compared to compression tasks, these two models focus more on improving the recognition accuracy of the models. The model proposed by Chen et al. [19] and TAKD [21] utilizes distillation strategies to facilitate cognitive transfer from complex networks to simpler ones.

In theory, different from natural images with simple targets, remote sensing data with wide-area views introduce a complex representation. Specifically, the targets related to categories

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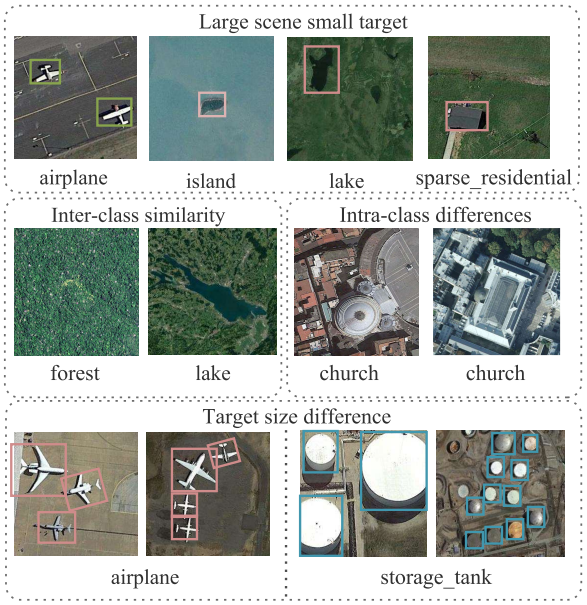


Fig. 1. Typical representation of complex remote sensing data expression. The first line depicts remote sensing data of large scenes and small targets, and the second line presents remote sensing data characterized by significant intraclass variations and minimal interclass differences. The third line is remote sensing data that reflect diverse changes in target size.

represent only a small portion of the complex scene, as illustrated in line 1 of Fig. 1. Similar scenes across different categories, as well as diverse scenes within the same category, are depicted in line 2 of Fig. 1. The significant variation in target size is highlighted in line 3 of Fig. 1. The complexity of the aforementioned scenarios places greater demands on the distillation strategy. How to transfer the recognition ability of teacher networks for complex scenes to student networks is a challenging issue.

The knowledge distillation model [19] does not take into account the remote sensing data complexity and directly applies the existing distillation method [22]. This approach is likely to produce suboptimal results that deviate from the optimal outcomes. Considering the complex representation of remote sensing data, the TAKD model [21] focuses on the recognition and transfer of target features to simplify remote sensing data interpretation models. Its focus is on the target itself within complex scenes, without exploring the underlying reasons for the scene's complexity. The method of transferring only the target features may hinder the student network's ability to effectively recognize environmental features, thereby impacting its generalization capabilities.

Based on the complexity of remote sensing data and existing studies, a novel knowledge distillation model is designed and applied to obtain lightweight neural networks with efficient processing capabilities for remote sensing data. The purpose of designing this new knowledge distillation model is to transfer the robust generalization capabilities of the teacher network, which excels in complex scenarios, to a simpler student network. This transfer enables the student network to develop approximation and fitting abilities that are unattainable through direct training and traditional distillation methods. Considering the complexity of remote sensing data representation, this knowledge distillation model incorporates a multiscale contrastive learning supervision module (MCSM) and a transformation consistency supervision module (TCSM). The MCSM enables student networks to identify similarities within the same category while also capturing diversity across different categories. Meanwhile, the TCSM assists student networks in learning target features of varying sizes.

The main contributions of the proposed method are given as follows.

1) A novel multiscale contrastive learning knowledge distillation model has been proposed. This model aims to transfer the generalization performance of a complex network to a simpler, lightweight network by introducing new constraints to the knowledge distillation process.
2) A MCSM has been proposed. This module offers additional guidance for the distillation process. Multiscale information is extracted and fused to enrich the guidance during distillation. Furthermore, a contrastive learning mechanism has been implemented to enhance the transfer of category generalization capabilities to the student network.
3) A TCSM has been designed to guide the feature learning of the student network. The distillation module effectively transfers the cognitive abilities of teacher networks across various target sizes to the student network, thereby enhancing the student network's generalize capacity in complex scenarios.

The rest of the article is organized as follows. In Section II, the works related to our proposed model are briefly summarized. In Section IV, the proposed method, execution process, and experimental results are introduced. In Section V, the effectiveness of the proposed method through comparative and ablation experiments is verified. A brief summary is made in Section VI.

II. Relative Works

A. Remote Sensing Scene Classification

Remote sensing scene classification entails categorizing remote sensing data into predefined groups, thereby enhancing the understanding of various data scenarios. Historically, the interpretation and classification of remote sensing data have predominantly depended on traditional methods, which involved the manual selection of features [23], [24], [25], [26], [27]. The development of deep neural networks, particularly CNNs and Transformers, has resulted in groundbreaking advancements in the classification of remote sensing data [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38].

Obtaining effective feature representations of remote sensing data is crucial for achieving high-performance interpretation in deep neural networks. Next, we will summarize the existing feature extraction methods from three perspectives: feature fusion, network fusion, and novel learning strategies. The feature fusion methods primarily consist of multilayer feature fusion and multisource feature fusion. Inspired by the feature pyramid structure, the multilayer feature fusion model enhances the feature representation extracted by the network

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by aggregating features from both the middle layer and the final layer [39], [40], [41], [42]. Multisource feature fusion involves extracting features from data originating from diverse sources and providing multiangle feature representations for classifiers, thereby enhancing performance [28], [43].

Network fusion improves model recognition performance by integrating the cognitive capabilities of multiple network branches. Specifically, ACCNN [29] designs classification branches and an attribute branch to extract discriminative features and category attributes, respectively, thereby enhancing the guidance for classifiers. MBLANet [44] incorporates a channel attention module and a local spatial attention module (LSAM) to capture both channel and spatial attention information. In addition, a twin network structure is designed to analyze input data from both global and local perspectives [11]. Subsequently, a dual-branch attention network is proposed to enhance hyperspectral remote sensing image classification tasks [45]. MGSNet [46] employs multiple network branches to improve the network's ability to recognize foreground and background elements. DFAGCN [47] effectively leverages the features extracted by both graph neural networks and CNNs.

In addition, methods for designing new learning strategies within the learning processes of deep neural networks have been proposed, achieving optimal results in remote sensing data interpretation tasks. D-CNNs [48] incorporate a metric learning-based regularization mechanism into the learning process of deep neural networks, providing positive guidance for the extraction of discriminative features. Meanwhile, contrastive learning strategies have also been employed to assist deep neural networks in identifying effective discriminative features in complex remote sensing data scenarios [49], [50].

The aforementioned methods involve the design of additional feature extraction schemes or constraint strategies within the fundamental network framework. Compared to basic network frameworks, these new solutions and strategies impose additional computational and storage demands. Achieving high-performance interpretations while minimizing hardware resource usage is the objective for deep neural networks [51].

B. Knowledge Distillation-Based Remote Sensing Scene Classification

Knowledge distillation is a learning method that transfers knowledge from a teacher network model to a student network model by minimizing the discrepancies between their predictions [22]. This method is frequently employed for model compression, where the student network is a lightweight alternative to the teacher network [52], [53], [54], as well as for model performance fusion, in which multiple teacher networks transfer knowledge to a single student network [55]. Currently, the knowledge distillation learning method is extensively utilized in classification, segmentation, and detection tasks based on natural image data [56], [57], [58].

Recently, research on the application of knowledge distillation to remote sensing data classification tasks has made significant progress. Currently, knowledge distillation-based learning methods in the field of remote sensing data classification primarily fall into two categories: self-distillation and mutual distillation. The latter achieves model compression and addresses the issue of insufficient training data by transferring knowledge between two distinct network models.

VKT [18] employs self-distillation as a supervisory method to enhance the network's classification capabilities for remote sensing datasets. Instead of relying solely on one-hot encoded labels for categorical guidance, it also incorporates class entanglement information as an additional supervisory input. This approach facilitates the student network's learning process, thereby improving its understanding of interclass correlations. In remote sensing data classification tasks, a series of knowledge distillation-based methods have been proposed and implemented to achieve a more efficient network that enhances classification accuracy while balancing performance with computational requirements. The model proposed by Xu et al. [20] transfers the features of the Transformer, which account for global information, to the CNN network through knowledge distillation, thereby balancing experimental accuracy with computational demands. To enhance the student network's ability to capture essential local features, the model introduced by Li et al. [59] combines distillation with spatial and spectral attention mechanisms, facilitating guidance from the teacher network to the student network. To alleviate the substantial performance difference between complex teacher networks and lightweight student networks, Liu et al. [60] propose a novel intermediate network model designed to facilitate the entire distillation process through a two-step distillation approach. This model assists in the learning process of the student network. In addition to the aforementioned models, MS2RGB-KD [61] proposes a method to distill the knowledge from a trained teacher network model, which operates on multispectral data, into a student network that accepts only RGB images as input. This approach allows the rich cognitive capabilities of the teacher network to enhance the learning process of the student network.

The purpose of these models is to improve the classification accuracy of the original model through a knowledge distillation strategy. Models such as those presented in [19] and [21] employ knowledge distillation to improve the performance of portable student networks, achieving better results compared to direct training. These methods do not thoroughly account for the complex representation of remote sensing data; they primarily rely on conventional distillation models or concentrate on the transfer of target features. The above distillation methods did not incorporate guidance based on complex scene cognition for student networks, and they are unable to effectively transfer the generalization capabilities of teacher networks to student networks.

III. Proposed Method

A. Preliminary Knowledge

Knowledge distillation facilitates the transfer of knowledge from a teacher network $F_t(\cdot)$ to a student network $F_s(\cdot)$. The conventional knowledge distillation is represented by the following equation:

$$\mathcal{L}_{\mathrm{KD}} = \min_{F_s(\cdot)} D_{\mathrm{kl}}\left(F_t(x), F_s(x)\right) \tag{1}$$

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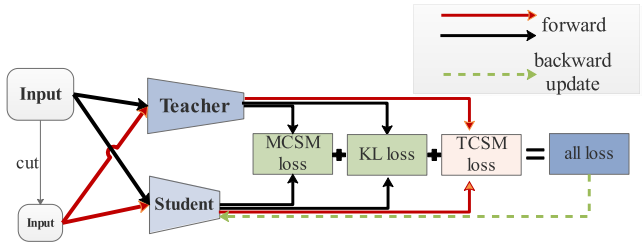


Fig. 2. Simplified representation of our proposed knowledge distillation model. The guidance module of this distillation model primarily consists of three components: a conventional supervision module based on KL divergence guidance, an MCSM with a black line, and a TCSM with a red line.

where x is the original training dataset, and $F_t(\cdot)$ and $F_s(\cdot)$ denote the output probability distributions of the teacher and student networks, respectively. D_{kl} refers to the Kullback–Leibler (KL) divergence, which is utilized to quantify the similarity between two distinct probability distributions. This equation is a commonly used target function in conventional knowledge distillation models.

B. Method Overview

The conventional knowledge distillation model described above employs the maximum KL divergence between the output logits of the teacher and student networks to achieve distillation, and this method only depends on the divergence of logits to provide learning guidance from the teacher network to the student network. In this article, a novel knowledge distillation model is proposed. This model introduces new constraints to the conventional knowledge distillation process, facilitating the transfer of the teacher network's cognitive abilities in complex scenes to the student network. The constraints of this novel distillation model consist of three main components: a conventional supervision component that utilizes KL divergence, an MCSM, and a TCSM. The design purpose of the MCSM and TCSM is to transfer the cognitive capabilities of the teacher network to the student network in complex scenarios, including intricate data scenarios, interclass similarity, intraclass diversity, and significant size variation. This transfer aims to enhance the generalization ability and overall cognitive performance of the student network in complex remote sensing scenarios.

As illustrated in Fig. 2, the objective function for this distillation model consists of three components: the MCSM loss derived from the MCSM, the TCSM loss obtained from the TCSM, and the KL divergence loss from conventional distillation, as defined in (1). The overall loss function of the new distillation model is presented as follows:

$$\mathcal{L}_{\text{all}} = \beta_1 \mathcal{L}_{\text{MCSM}} + \beta_2 \mathcal{L}_{\text{TCSM}} + \beta_3 \mathcal{L}_{\text{KD}}. \tag{2}$$

To provide a detailed introduction to the new supervision modules, we present the execution details in Fig. 3, followed by a comprehensive overview of the proposed distillation model.

C. Multiscale Contrastive Learning Supervision Module

In Fig. 3, the MCSM, associated with the MCSM loss, is designed to more effectively transfer the generalized cognitive abilities of the teacher network for complex scenes to the student network. In this module, a multiscale feature fusion strategy is proposed. The fused features, which contain rich information, are regarded as guiding information in the MCSM. In addition, this module introduces a contrastive distillation strategy that utilizes the fused features of teacher networks to assist student networks in identifying complex features of remote sensing categories. Category-based contrastive learning allows samples from the same category to exhibit similar representations in the feature space, while samples from different categories demonstrate significantly distinct representations. Therefore, contrastive learning can assist teacher networks in transferring their generalized cognitive abilities related to category information—including interclass differences and intraclass similarities—to student networks. This process enables student networks to acquire more discriminative category information.

To optimize the utilization of the discriminative category features extracted by the teacher network, this module extracts and integrates features of varying scales learned by the teacher network for guidance. According to the principles of deep neural networks, deeper layers are capable of responding to high-level semantic features. However, the limited size of high-level features results in insufficient geometric information, which hampers the recognition of complex scenes. In contrast, shallow networks contain rich geometric information; they lack sufficient semantic features, thereby hindering effective scene classification. Therefore, capturing multiscale contextual information and utilizing a combination of coarse-to-fine features enhance the network's ability to learn and understand data features. This issue is particularly evident in the recognition and interpretation of remote sensing images. Consequently, we extract and utilize the fused information from the multiscale features of the teacher network to guide the distillation process.

In the process of extracting and fusing multiscale features, the principles of the feature pyramid and the self-attention mechanism are employed. Once the teacher network has acquired the multiscale feature representation, the corresponding features at each scale serve as References for performing contrastive learning. This approach enhances the student network's comprehension of intraclass similarities and interclass diversities. The execution process of this module is illustrated by the solid black line in Fig. 3.

As illustrated in Fig. 3, when performing supervised learning using the MCSM, the original image serves as input to the distillation model, and different hierarchical features C_1, C_2, C_3, and C_4 are extracted sequentially from the trained teacher network. Following the multiscale feature extraction and fusion scheme depicted in the figure, the highest-level feature C_4 is upsampled using bilinear interpolation and then cascaded with C_3 to produce the cascaded feature C_3'. The cascaded feature is further upsampled to generate feature V. C_3' is upsampled and cascaded with C_2 to create the cascaded feature C_2', which is designated as feature K. After upsampling C_2' in a similar manner and cascading it with C_1 to obtain C_1', C_1' is downsampled to obtain feature Q.

Inspired by the EMTCAL model [42] and the multiscale feature fusion model [62], this module utilizes an

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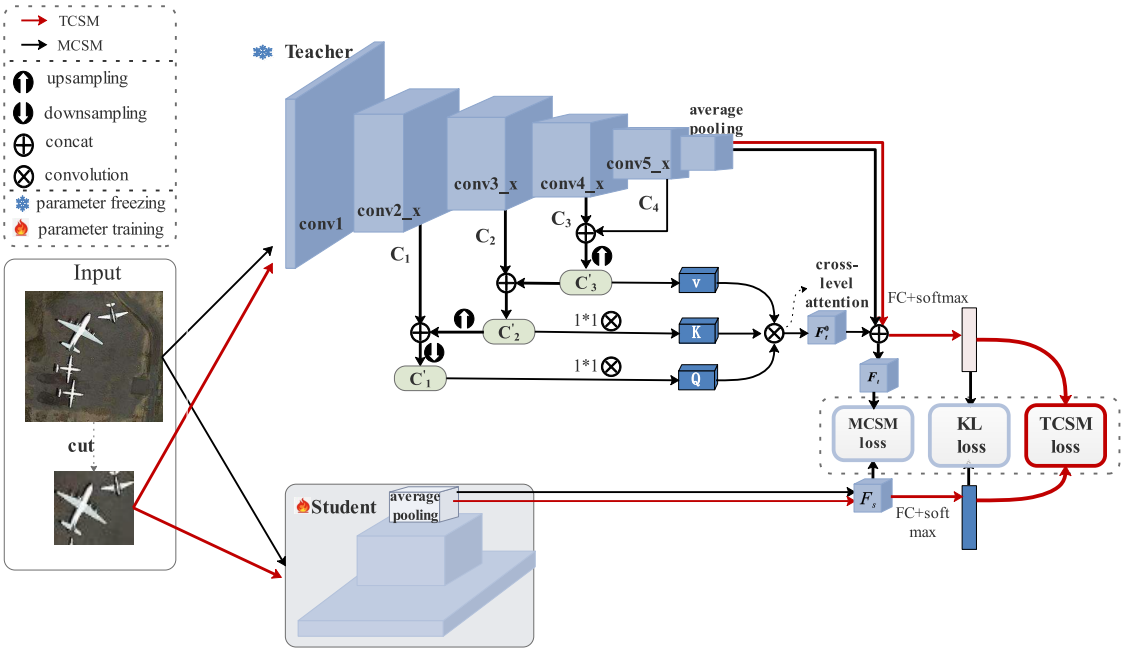


Fig. 3. Detailed representation of our proposed knowledge distillation model is presented. The guidance module of this distillation model primarily comprises three components: a conventional supervision module based on KL divergence guidance, an MCSM, and a TCSM. The black solid line illustrates the execution process of the MCSM module, while the red solid line depicts the execution process of the TCSM module.

attention mechanism to achieve the fusion of multiscale features, referred to as cross-level attention fusion. From the description above, it is evident that different levels and scales of features V, K, and Q are extracted from the trained teacher network. These three features are fused into the spatial multiscale self-attention feature F_t^0 through the spatial self-attention mechanism, which employs a dot-product operation. The fusion method, based on the execution of the self-attention mechanism, is outlined as follows:

$$F_t^0 = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q * K^T}{\sqrt{d}}\right) * V \tag{3}$$

where $(d)^{1/2}$ is the scaling factor for adjusting the size of the numerical value. To align the dimensions of V, Q, and K, a 1×1 convolutional layer is applied separately to K and Q, enabling them to attain the same dimension as V.

After global average pooling and normalization, this fused feature is cascaded with the final average pooling layer feature of the teacher network to produce the ultimate feature representation, denoted as F_t. This multiscale fusion-based feature representation serves as the core representation that the teacher network uses to guide students through contrastive learning in the MCSM. Analogously, the penultimate features of the student network are extracted to represent the feature vector F_s for the input data. Since contrastive learning is trained by evaluating the similarities and differences between samples, this article employs simple cosine similarity as the metric for measuring the feature similarity between data x_1 and x_2

$$\text{sim}(F(x_1), F(x_2)) = \frac{< F(x_1), F(x_2) >}{\|F(x_1)\| * \|F(x_2)\|} \tag{4}$$

where x_1 and x_2 are any two different data points.

As illustrated in Fig. 3, the fully connected (FC) layer of the trained teacher network did not participate in the training process previously; it solely provided multiscale features for distillation. However, due to the extraction of these multiscale features, the FC layer needs to be updated. During this update process, the 1×1 convolution operations used to unify the dimensions of V, K, and Q are also optimized and trained simultaneously. The objective function of this process is

$$\mathcal{L}_{\text{CE}} = \text{CrossEntropy}(y_t, \text{FC}(F_t)) \tag{5}$$

where y_t is the category label and FC$(\cdot)$ is the FC layer.

To provide cognitive guidance to students regarding interclass differences and intraclass similarities, we utilize the multiscale feature F_t of the teacher network as the foundational representation. This serves to establish contrastive optimization objectives for the student network. To accomplish this, we employ the following objective function to extract the multiscale features F_t:

$$\mathcal{L}_{T-\text{MSC}} = -\mathbb{E}_{x \in X} \log\left[\frac{\sum_i \sum_p^P \exp\left(\text{sim}\left(F_t(x_i), F_t\left(x_p^+\right)\right)\right)}{\sum_i \sum_n^N \exp\left(\text{sim}\left(F_t(x_i), F_t\left(x_n^-\right)\right)\right)}\right] \tag{6}$$

where x_p^+ is the pth data, which has the same category with x_i, and x_n^- is the nth data, which has the different category with x_i. During each training iteration, X is the input dataset and each instance $x_i \in X$, $x_p^+ \in X$, and $x_n^- \in X$. For each x_i, the positive samples are all sample data with the same category as x_i except for x_i, and its positive sample size is P, $p \in P$. Its corresponding negative samples are all sample data with different categories as x_i, and its negative sample size is N, $n \in N$. During the training of the student network, we provide feature guidance for the penultimate layer's second features to

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ensure that it learns the teacher network's overall robustness concerning intraclass similarity and interclass differences

$$\mathcal{L}_{T-S} = \sum_{x=1}\sum_{j=1}\log\left[\exp\left(\mathrm{sim}\left(F_t\left(x_i\right),F_t\left(x_j\right)\right)\right) - \exp\left(\mathrm{sim}\left(F_s\left(x_i\right),F_s\left(x_j\right)\right)\right)\right]^2. \quad (7)$$

In addition to guidance from teacher networks, student networks also enhance their understanding of interclass differences by contractive learning

$$\mathcal{L}_{S-\mathrm{MSC}} = -\mathbb{E}_{x\in X}\log\left[\frac{\sum_i\sum_p^P\exp\left(\mathrm{sim}\left(F_s\left(x_i\right),F_s\left(x_p^+\right)\right)\right)}{\sum_i\sum_n^N\exp\left(\mathrm{sim}\left(F_s\left(x_i\right),F_s\left(x_n^-\right)\right)\right)}\right]. \quad (8)$$

Therefore, the total loss of the MCSM module is

$$\mathcal{L}_{\mathrm{MCSM}} = \mathcal{L}_{T-\mathrm{MSC}} + \mathcal{L}_{T-S} + \mathcal{L}_{S-\mathrm{MSC}} + \mathcal{L}_{\mathrm{CE}}. \quad (9)$$

D. Transformation Consistency Supervision Module

The red lines in Fig. 3 represent a simplified depiction of the input of TCSM, the feature transmission path, and the associated loss. The purpose of this module is to achieve consistent cognition between the student network and the teacher network, regardless of the varying sizes of the target transformations. The execution process of this module is illustrated by the solid red line in the figure. The TCSM enhances the cognitive guidance provided by the teacher network to the student network by maximizing the consistency of the output between the two networks in various scale objectives. In remote sensing data scene classification tasks, the challenges extend beyond complex scenes. Issues such as large scene areas and small targets also present significant obstacles to enhancing the performance of lightweight models. This is exemplified in Fig. 1, which demonstrates that in certain scenarios, the target area occupies less than half of the entire scene. Consequently, this issue can cause lightweight networks to misinterpret the focus of category features, erroneously extracting features from other areas.

To address this issue, the TCSM extracts half or a quarter of the original image data containing the targets and resizes it to match the dimensions of the original image for the input of the network. By maximizing the consistent recognition of the same input between the teacher and student networks, the TCSM guides the student network in learning to recognize remote sensing data targets of varying sizes.

The purpose of this module is to ensure consistent recognition of inputs at various scales between teacher and student networks. Consequently, the loss function for this module is given as follows:

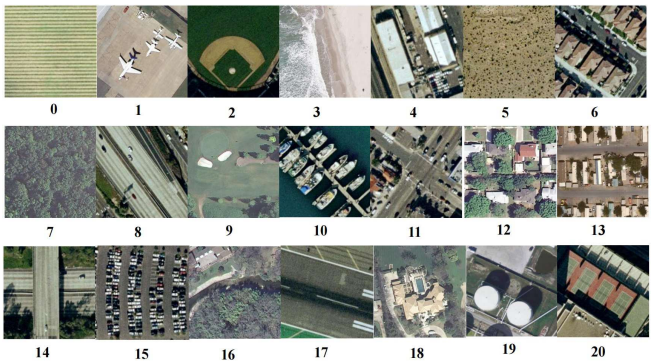
$$\mathcal{L}_{TCSM} = \min_{F_s(\cdot)} D_{\mathrm{kl}}\left(F_t\left(\hat{x}\right)\left(F_s\left(\hat{x}\right)\right)\right) \quad (10)$$

where $\hat{x}$ is the new data after cutting and resizing based on the original input data and D_{kl} is the KL divergence.

IV. Experiments

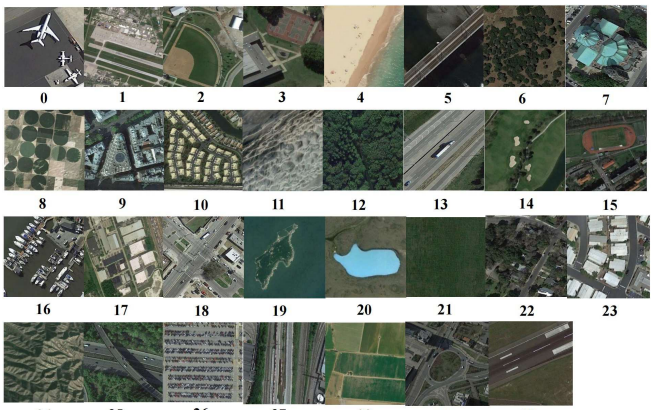
A. Dataset and Evaluation Metrics

To verify the effectiveness of the distillation model proposed in this article, we conducted a series of comparative and ablation experiments on four benchmark remote sensing datasets,



0:Agricultural,1:airplane,2:baseballdiamond,3:beach,4:buildings,5:chaparral,6:denseresidential, 7:forest,8:freeway,9:golfcourse,10:harbor,11:intersection,12:mediumresidential, 13:mobilehomepark,14:overpass,15:parkinglot,16:river,17:runway,18:sparseresidential, 19:storagetanks,20:tenniscourt

(a)



0:Airplane,1:airport,2:baseball_diamond,3:basketball_court,4:beach,5:bridge,6:chaparral,7:church, 8:circular_farmland,9:commercial_area,10:dense_residential,11:desert,12:forest,13:freeway, 14:golf_course,15:ground_track_field,16:harbor,17:industrial_area,18:intersection,19:islad, 20:lake,21:meadow,22:medium_residential,23:mobile_home_park,24:mountain,25:overpass, 26:parking_lot,27: railway ,28: rectangular_ farmland,29: roundabout,30:runway

(b)

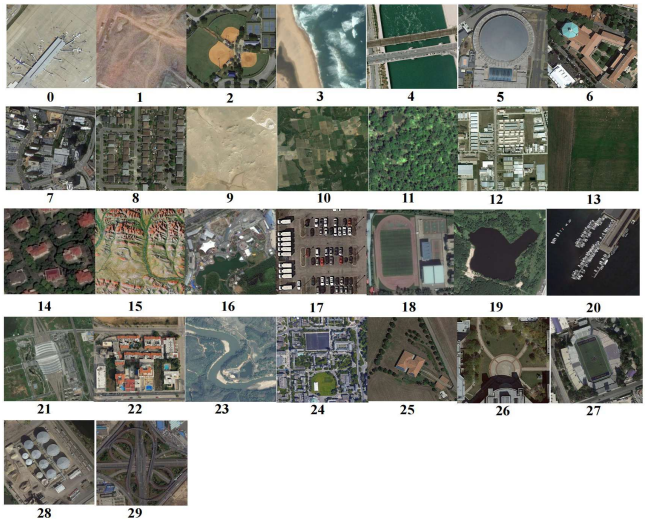
Fig. 4. Simplified representations of the UC Merced dataset and the OPTIMAL-31 dataset. (a) Representative data and category names of the UC Merch dataset. (b) Representative data and category names of the OPTIMAL-31 dataset.

including NWPU-RESISC45 [7], AID [63], OPTIMAL-31 [64], and UC-Merced [65]. To avoid the randomness of the results, we randomly selected the training data according to a specific proportion and conducted five repeated experiments, reporting the mean and standard deviation.

The UC Merced dataset was published by the University of California, Merced. It contains 2100 remote sensing scene images across 21 distinct categories, with each category comprising 100 scene images with a size of 256 × 256 pixels. Some scenes and their corresponding category labels from this dataset are illustrated in Fig. 4(a). To ensure the fairness of the comparative experiment, the proportion of training data that we selected is consistent with that of the existing remote sensing data distillation model [18], [21], [42], specifically 50% and 80% of the total dataset.

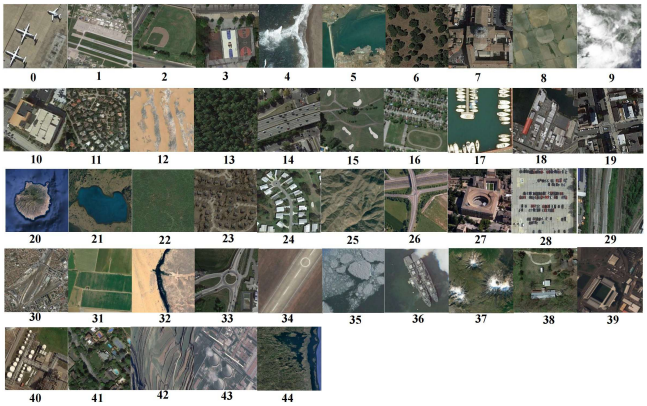
The OPTIMAL-31 dataset, published by Northwestern Polytechnical University, comprises 31 scene categories and each containing 60 images with a resolution of 256 × 256 pixels. Fig. 4(b) illustrates examples of several scenes along

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0:airport,1:bareLand,2:basebaUField,3:beach,4:bridge,5:center,6:church,7:commercial, 8:denseResidential,9:desert,10:farmland,ll:forest,12:industria1,13:meadow 14:mediumResidential, 15:mountain,16:park,17:parking,18:playground,19:pond,20:port,21:railwayStation,22:resort, 23:river,24:school,25:sparseResidential,26:square,27:stadium,28:storageTanks,29:viaduct

(a)



0:airplane,1:airport,2:baseball_diamond,3:basketball_court,4:beach,5:bridge,6:chaparral,7:church, 8:circular_frmland,9:cloud,10:commercial_area,11:dense_residential,12:desert,13:frest,14:feeway, 15:golf_course,16:ground_track_field,7:harbor,18:industrial_area,19:intersection,20:island, 21:lake,22:meadow,23:medium_residential,24:mobile_home_park,25:mountain,26:overpass, 27:palace,28:parking_lot,29:railway,30:railway_station,31:rectangular_frmland,32:river, 33:roundabout,34:runway,35:sea_ice,36:ship,37:snowberg,38:sparse_residential,39:stadium, 40:storage_tank,41:tennis_court,42:terrace,43:thermal_power_station,44:wetland

(b)

Fig. 5. Simplified representations of the AID dataset and the NWPU-RESISC45 dataset. (a) Representative data and category names of the AID dataset. (b) Representative data and category names of the NWPU-RESISC45 dataset.

with their corresponding category labels from this dataset. In accordance with the existing model, we designated 80% of the total data as the training set.

The AID dataset, published by Wuhan University, contains 10 000 remote sensing scene images. This dataset includes 30 scene categories, with each category containing between 220 and 240 images, each measuring 600 × 600 pixels. Examples of various scenes along with their corresponding category labels from this dataset are presented in Fig. 5(a). To compare with the existing models, we choose the data training proportion consistent with these models [18], [21], [42], [66], that is, 20% and 50%.

The NWPU-RESISC45 dataset was constructed by Northwestern Polytechnical University and contains 31 500 remote sensing scene images. This dataset comprises 45 scene categories, each category containing 700 images, with a size of 256 × 256. Some scenes and their corresponding category labels of this dataset are illustrated in Fig. 5(b). To ensure the fairness of the comparative experiment, we select 10% and 20% of the training data by following established methods [18], [21], [42].

TABLE I

NETWORK STRUCTURE AND PARAMETER COUNT OF ALL TEACHER AND STUDENT NETWORKS INVOLVED IN THE DISTILLATION EXPERIMENT

	Network	Params	TestTime
Teacher	ResNet50	42.5M	6.5s
	Resnet34	21.8M	4.7s
Student	GoogleNet	13.0M	1.6s
	MobilenetV2	2.3M	0.9s
	BinaryDNet-53	0.4M	0.7s
	ResNet18	11.7M	1.2s
	SqueezeNet1_0	0.7M	0.8s
	LSKNet-T	4.3M	1.1s

Similar to the existing model evaluation criteria for exploring remote sensing data classification tasks [4], [42], we selected overall accuracy (OA) as the standard to evaluate the performance of the distillation model. This evaluation criterion is defined as the ratio of correctly classified images to the total number of test images, which reflects the overall performance of a classification model. In addition, the ablation experiment and effectiveness analysis are conducted to validate the efficacy of the distillation model proposed in this article.

B. Experimental Settings

During the experiment, we selected ResNet50 and ResNet34 as the teacher networks. To transfer the performance of these complex models to lightweight networks and enhance the scene recognition capabilities of these networks, we utilize ResNet18, MobilenetV2, squeezenet1_0, GoogleNet, LSKNet [67], and BinaryDNet53 [68] as student networks for distillation training. To quantitatively demonstrate the lightweight nature of the selected student networks, the parameters and test times for an image on different networks are presented in Table I. Distillation experiments can be categorized into two scenarios: those that utilize pretrained models and those that do not. In all experimental results, the blackened data represent optimal accuracy. It is important to note that all the experiments have been conducted five times, and both the average value (Ave.) and the standard deviation (std.) are provided.

All experiments were conducted using PyTorch [69] on a workstation equipped with NVIDIA Titan V100 GPUs. The partial experiments conducted on the UC Merced dataset do not utilize pretrained models. In these experiments, the training of both the teacher and student networks is conducted without loading the pretrained model. We utilize the Adam optimizer to train the student network for 200 epochs. The initial learning rate is set to 0.01, and we utilize a cosine decay learning rate scheduler with a linear warm-up. The experiments conducted

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TABLE II
COMPARATIVE EXPERIMENTAL RESULTS BASED ON THE UC MERCED DATASET

Model	Training	GFLOPs	Training rote	
			50%	80%
T (Resnet34)	direct	10.99G	82.37± 0.21	98.39±0.20
S_1 (Resnet18)	direct	5.44G	81.01±0.17	97.08±0.37
	KD	9.10G	81.73± 0.25	97.41 ±0.08
	Ours	9.13G	**82.24± 0.21**	**98.22±0.15**
S_2 (Mobilenetv2)	direct	0.90G	78.65±0.36	95.23±0.27
	KD	4.56G	79.12±0.22	95.57±0.41
	Ours	4.57G	**81.59±0.17**	**96.45±0.13**
S_3 (Googlenet)	direct	4.49G	79.03±0.15	94.31±0.40
	KD	8.15G	79.65±0.20	95.02±0.55
	Ours	8.17G	**81.23±0.28**	**96.69±0.18**
S_4 (LSKNet-T)	direct	4.29G	81.21±0.12	97.17±0.10
	KD	7.95G	81.48±0.06	97.66±0.09
	Ours	7.97G	**82.16±0.15**	**98.02±0.12**
S_5 (Squeezenet1_0)	direct	2.47G	74.23±0.25	92.85±0.18
	KD	6.13G	76.07±0.41	93.62±0.11
	Ours	6.15G	**78.79±0.20**	**95.47±0.26**
S_6 (BinaryDNet53)	direct	2.12G	78.95±0.17	94.71±0.20
	KD	5.78G	79.34±0.05	95.69±0.50
	Ours	5.80G	**81.25±0.19**	**97.14±0.30**

TABLE III
COMPARATIVE EXPERIMENTAL RESULTS BASED ON THE AID DATASET

Model	Training	GFLOPs	Training rote	
			20%	50%
T (Resnet50)	direct	12.27G	95.57± 0.11	97.08±0.12
S_1 (Resnet18)	direct	5.44G	92.31±0.12	95.31 ±0.31
	KD	9.53G	93.70±0.22	95.83 ± 0.41
	Ours	9.55G	**94.62 ± 0.11**	**96.34 ± 0.13**
S_2 (Mobilenetv2)	direct	0.90G	92.30 ± 0.15	95.38± 0.11
	KD	4.99G	93.17 ±0.21	95.61± 0.17
	Ours	5.02G	**95.12±0.25**	**96.28± 0.15**
S_3 (Googlenet)	direct	4.49G	93.12±0.31	95.64±0.18
	KD	8.58G	93.85±0.40	95.82±0.19
	Ours	8.61G	**95.35± 0.24**	**96.23±0.13**
S_4 (LSKNet-T)	direct	4.29G	93.13±0.12	96.80±0.09
	KD	8.38G	93.65±0.22	96.84±0.12
	Ours	8.40G	**95.12±0.12**	**97.11±0.13**
S_5 (Squeezenet1_0)	direct	2.47G	88.20±0.17	92.46±0.08
	KD	6.56G	89.42±0.21	93.15±0.31
	Ours	6.59G	**91.79±0.27**	**94.44±0.19**

TABLE IV
COMPARATIVE EXPERIMENTAL RESULTS BASED ON THE NWPU-RESISC45 DATASET

Model	Training	GFLOPs	Training rote	
			10%	20%
T (Resnet50)	direct	12.27G	93.40± 0.20	95.43±0.11
S_1 (Resnet18)	direct	5.44G	88.91±0.23	91.77±0.18
	KD	9.53G	90.51±0.30	92.38± 0.19
	Ours	9.55G	**91.45±0.10**	**93.68±0.11**
S_2 (Mobilenetv2)	direct	0.90G	89.59± 0.13	92.38±0.17
	KD	4.99G	90.15±0.22	93.33±0.11
	Ours	5.02G	**91.60±0.26**	**93.84±0.19**
S_3 (Googlenet)	direct	4.49G	85.40± 0.25	91.93±0.16
	KD	8.58G	86.72± 0.12	92.40±0.21
	Ours	8.61G	**88.71± 0.17**	**93.20± 0.13**
S_4 (LSKNet-T)	direct	4.29G	90.35±0.22	94.07±0.16
	KD	8.38G	91.07±0.17	94.25±0.09
	Ours	8.40G	**93.29±0.10**	**95.30±0.12**
S_5 (Squeezenet1_0)	direct	2.47G	85.16±0.27	88.81±0.31
	KD	6.56G	86.26± 0.11	90.46 ±0.13
	Ours	6.59G	**88.23±0.10**	**92.50 ± 0.10**

on the NWPU-RESISC45, AID, OPTIMAL-31, and partial experiments on the UC Merced datasets were all based on pretrained models. With the exception of the fully connected layer, the training of both the teacher and student networks utilized pretrained models from the ImageNet dataset.

We utilized the Adam optimizer to train the student networks for 200 epochs. The initial learning rate was set to 0.0001, and we implemented a cosine decay learning rate scheduler with a linear warm-up. The batch size is set to 32, and all input scenes are resized into 224 × 224. All training data ratios are derived from existing research [18], [20], [33], [42], [70]. β_1, β_2, and β_3 in (2) are all set to 1. In the ablation experiment, we will conduct a further analysis of the numerical settings for β_1 and β_2.

In the MCSM, bilinear interpolation upsampling is employed for multiscale feature extraction and fusion, utilizing the upsampling function from PyTorch version 1.7.0. The target size for each upsampling operation is aligned with the feature that is to be cascaded.

C. Comparative Experiments

To evaluate the effectiveness of our proposed distillation model, we divided the comparative experiments into three distinct sections. The first section involves accuracy comparisons with direct training and conventional distillation methods [22]. The second section compares our model with existing distillation models utilized for remote sensing data classification. The third section evaluates the classification accuracy of ResNet34 utilizing our proposed distillation method, in comparison to several enhanced versions of ResNet34. The data highlighted in black indicate the optimal accuracy.

1) Experiment 1: Compared to Direct Training: Tables II–V show the experimental results of the first section in the comparative experiment. In the results, "direct" represents the conventional training [16]. The "KD" refers to the conventional distillation [22]. The term "ours" in the tables refers to the method proposed in this article. Table II presents the experimental results based on the UC Merced dataset, which compares our proposed method with direct training and conventional distillation methods. The experimental results are not based on pretrained models. The results presented in the table indicate that our proposed

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TABLE V
COMPARATIVE EXPERIMENTAL RESULTS BASED ON THE OPTIMAL-31 DATASET

Model	Training	GFLOPs	Training rote
			80%
T (Resnet50)	direct	12.27G	97.31± 0.20
S_1 (Resnet18)	direct	5.44G	94.48± 0.13
	KD	9.53G	94.91± 0.18
	Ours	9.55G	**96.27± 0.17**
S_2 (Mobilenetv2)	direct	0.90G	94.81± 0.17
	KD	4.99G	95.67±0.11
	Ours	5.02G	**96.45± 0.12**
S_3 (Googlenet)	direct	4.49G	94.62± 0.27
	KD	8.58G	95.13±0.08
	Ours	8.61G	**96.27± 0.10**
S_4 (LSKNet-T)	direct	4.29G	95.35±0.32
	KD	8.38G	96.02±0.07
	Ours	8.40G	**97.13±0.20**
S_5 (Squeezenet1_0)	direct	2.47G	89.25±0.31
	KD	6.56G	91.02± 0.18
	Ours	6.59G	**92.29± 0.15**

multiscale contrastive distillation model exhibits superior performance. When the training dataset size is 50%, our distillation method achieved an accuracy improvement ranging from 2% to 4.5% compared to direct training. In comparison to the vanilla distillation, our method realized an accuracy increase of 1.5%–2.7%. When the training dataset size is increased to 80%, our proposed distillation method achieved an accuracy enhancement of 1.2%–2.6% over direct training. Compared to vanilla distillation methods, our method demonstrated an accuracy improvement of 0.9%–1.8%. From the experimental results presented above, it is evident that the proposed distillation model outperformed both direct training and vanilla distillation. Furthermore, this advantage is particularly pronounced in lighter student networks.

Given the limited data volume, all subsequent distillation experiments were conducted using models that had been pretrained on the ImageNet dataset. Because of the difference in the number of sample categories, only the portion of the pretrained parameters preceding the fully connected layer was utilized. Table III presents the experimental results based on the AID dataset, comparing our proposed method with direct training and vanilla distillation methods.

From Table III, it is evident that the student network trained using our proposed distillation method demonstrates superior classification performance compared to both direct training and vanilla distillation. When the training data ratio is 20%, our distillation method increases the accuracy of the student network by approximately 3.0% compared to direct training and by an additional 2.0% over vanilla distillation. With a training data ratio of 50%, our method enhances accuracy by about 0.3%–2.0% over direct training and by 0.3%–1.3% more than vanilla distillation.

Similar to the operations conducted on the first two datasets, Tables IV and V present the experimental results on the NWPU-RESISC45 and OPTIMAL-31 datasets, respectively. The classification accuracy achieved by our proposed distillation model is significantly superior to that obtained through direct training and the vanilla distillation method. This finding aligns with the conclusions obtained on the two datasets mentioned earlier. On the NWPU-RESISC45 dataset, when the training data ratio is set at 10%, our distillation method enhances accuracy by 2.5%–3.3% compared to direct training. In addition, compared to vanilla distillation methods, our approach yields an accuracy improvement of 0.9%–2.0%. When the training data ratio is 20%, our distillation method improves accuracy by 0.53%–3.50% compared to direct training. In comparison to traditional distillation methods, our approach enhances accuracy by 0.35%–1.90%. To ensure a fair comparison, we set the training ratio for the OPTIMAL-3 dataset to 80%. The classification accuracies of the student network achieved using the proposed distillation method are 1.6%–2.0% higher than those obtained through direct training. In comparison to vanilla distillation, the increase in accuracy ranges from 0.8% to 1.3%.

2) Experiment 2: Compared to Existing Distillation Models: The comparative experiment in the first section confirmed the superior performance of our proposed distillation model when transferring knowledge from ResNet34 and ResNet50 to several typical lightweight networks. To further validate the effectiveness of the proposed method, we compared its performance with that of existing distillation models designed for classifying remote sensing data, such as VSD-ResNet34 [18], TAKD [21], and ET-GSNet [20], and a series of classic distillation models [52], [71], [72], [73], [74], [75], [76]. In these comparative experiments, ResNet101 and ResNet50 served as the teacher networks, while ResNet34 was utilized as the student network. The accuracy of the ResNet101 network in this experiment was achieved through 600 epochs of training with pretrained parameters from the ImageNet dataset. The initial learning rate is set to 0.001. The accuracies of ResNet101, used as the teacher network under a fixed proportion of the training dataset, are given as follows: AID (20%: 95.93 and 50%: 97.55), NWPU (10%: 94.10 and 20%: 95.62), UC Merced (50%: 99.44 and 80%: 99.89), and Optimal-31 (80%: 98.10).

In Table VI, "our-1" denotes the distillation results achieved using ResNet50 as the teacher network, whereas "our-2" indicates the distillation results obtained with ResNet101 as the teacher network. The results highlighted in bold indicate the highest accuracy, while the underlined results denote the second-highest accuracy. Based on the experimental results, we can conclude that our proposed distillation model has achieved an accuracy that is comparable to, or even exceeds, that of existing distillation models. Specifically, when employing ResNet101 as the teacher network, the majority of the distillation results obtained through the proposed method achieved the highest classification accuracy, with improvements ranging from approximately 0.07%–3.2%. However, there was a slight underperformance compared to VSD-ResNet34 when the training set comprised 20% of the AID dataset.

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TABLE VI

COMPARATIVE EXPERIMENTAL RESULTS OF OUR METHOD WITH OTHER DISTILLATION MODELS. "OUR-1" IS THE DISTILLATION RESULTS OF TAKING RESNET50 AS TEACHER NETWORK, AND "OUR-2" IS THE DISTILLATION RESULTS OF TAKING RESNET101 AS TEACHER NETWORK

Method	Param	GFLOPs	NWPU-RESISC45		UC Merced dataset		AID		OPTIMAL-31
			10%	20%	50%	80%	20%	50%	80%
FitNet	20.68M	50.80	76.58± 0.21	84.26±0.07	89.19± 0.12	92.14±0.17	81.55±0.12	88.52±0.15	-
RKD	21.8M	20.3	78.22± 0.10	86.69±0.22	92.74± 0.09	95.46±0.23	83.35±0.07	91.91±0.20	-
GiftKD	21.8M	19.81	77.04± 0.10	85.37±0.22	90.65± 0.09	92.96±0.23	82.67±0.07	89.16±0.20	-
AT	21.8M	21.91	83.01± 0.12	89.71±0.16	94.16± 0.15	96.09±0.10	86.50±0.21	93.52±0.16	-
DKD	21.3M	19.8	83.02±0.17	89.15±0.12	93.80± 0.11	96.42±0.05	88.21±0.14	93.33±0.09	-
ReviewKD	26.68M	32.38	86.51±0.05	91.07±0.10	95.35± 0.04	97.09±0.13	88.94±0.09	94.44±0.11	-
Target-aware	25.72M	30.51	85.62±0.05	91.21±0.10	95.94± 0.04	97.00±0.13	89.23±0.09	95.32±0.11	-
VSD-Resnet34	21.8M	16.43	92.13± 0.16	94.68±0.13	98.49± 0.18	99.67±0.18	**96.00±0.18**	97.28±0.14	96.95±0.17
TAKD	21.8M	9.12	89.01± 0.20	92.26±0.07	97.10± 0.10	98.41±0.19	92.20±0.13	95.35±0.21	-
ET-GSNet	21.8M	8.32	92.72±0.28	94.50±0.18	-	-	95.58±0.18	96.88±0.19	-
our-1	21.8M	15.09	92.89±0.13	95.10±0.10	98.71± 0.12	99.51± 0.13	95.37± 0.10	96.92± 0.14	97.13± 0.12
our-2	21.8M	18.79	**93.23±0.12**	**95.17± 0.12**	**99.32±0.15**	**99.74±0.21**	95.82±0.37	**97.31±0.17**	**97.61±0.22**

TABLE VII

COMPARATIVE EXPERIMENTAL RESULTS OF OUR METHOD AND THE IMPROVED MODEL BASED ON THE RESNET34 NETWORK. "OUR" IS THE DISTILLATION RESULTS OF THE PROPOSED DISTILLATION MODEL WITH THE RESNET101 AS THE TEACHER NETWORK

Method	Params	GFLOPs	NWPU-RESISC45		UC Merced dataset		AID	
			10%	20%	50%	80%	20%	50%
ACR-MLFF	-	-	90.01± 0.33	93.65±0.12	97.99± 0.26	99.37±0.18	92.73±0.12	95.66±0.33
EMTCAL	27.0 M	4.23	91.63±0.19	92.45±0.20	98.67±0.16	99.57±0.28	94.69±0.14	96.46±0.23
MF^2CNet	30.12M	2.42	92.07±0.22	93.85±0.27	98.76±0.16	99.52±0.25	95.54±0.17	97.02±0.28
our	21.8M	18.79	**93.23±0.12**	**95.17± 0.12**	**99.32±0.15**	**99.74±0.21**	**95.82±0.37**	**97.31±0.17**

When ResNet50 is used as the teacher network, the distillation model proposed in this article outperforms existing distillation models on both the NWPU-RESISC45 and OPTIMAL-31 datasets, achieving the second-highest classification accuracy, following the results of "our-2."

3) Experiment 3: Compared to Improved ResNet34: In addition to the comparative experiments mentioned above, we also compared the performance of the improved ResNet34. Specifically, we compared the performance of the ResNet34 model obtained through this distillation method with enhanced ResNet34 variants that have more parameters, such as ACR-MLFF [70], *MF*^{2}*CNet* [66], and EMTCAL [42].

In this comparative experiment, ResNet101 serves as the teacher network to guide the training of the ResNet34 network. The table displays the number of parameters for the comparative methods. To facilitate comparison with the existing remote sensing data classification methods mentioned earlier, this experiment employs pretrained models based on the ImageNet dataset. From Table VII, it is evident that the ResNet34 model, trained using the proposed distillation method outlined in this article, achieved the highest classification accuracy while maintaining the fewest parameters compared to the improved models with a greater number of parameters. In detail, the maximum improvement on the NWPU-RESISC45 dataset can reach 2.7%, and the maximum improvement on the UC Merced dataset can reach 1.4%.

The comparative experiments conducted above evaluated the comparison results of our proposed distillation method with direct training, vanilla distillation, existing improved distillation models, and improved ResNet34 models. The results of these experiments confirmed the effectiveness of our proposed distillation model.

D. Ablation Study

The comparative experiments conducted above have confirmed the effectiveness of the model proposed in this article. Next, we will evaluate the individual effects of the MCSM and TCSM through ablation experiments. To simplify the process, we selected the AID and UC Merced datasets for these experiments. Consistent with the previous comparative experiment, ResNet34 was chosen as the teacher model for the UC Merced dataset, while MobileNetV2 and Squeezenet1_0 were utilized as student networks for model distillation. For the AID dataset, ResNet50 was selected as the teacher network, with MobileNetV2, Mobilenetv2, and Sqeezenet1_0 being used as student networks for distillation.

Tables VIII and IX present ablation results of the MCSM and TCSM modules using the UC Merced and AID datasets,

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TABLE VIII

ABLATION EXPERIMENTAL RESULTS ON THE STUDENT OF MOBILENETV2 AND THE DATASET OF UC MERCED AND AID

Dataset	Ablation Module	Training rote	
		50%	80%
UC Merced	KD	79.12	95.57
	KD+TCSM	79.86(0.54↑)	95.92(0.35↑)
	KD+MCSM	81.01(1.89↑)	96.03(0.46↑)
	KD+TCSM+MCSM	81.59(2.47↑)	96.45(0.88↑)
Dataset	Ablation Module	Training rote	
		20%	50%
AID	KD	93.17	95.61
	KD+TCSM	93.58(0.41↑)	95.84(0.23↑)
	KD+MCSM	94.48(1.31↑)	96.02(0.41↑)
	KD+TCSM+MCSM	95.12(1.95↑)	96.28(0.68↑)

TABLE IX

ABLATION EXPERIMENTAL RESULTS ON THE STUDENT OF SQUEEZENET1_0 AND THE DATASET OF UC MERCED AND AID

Dataset	Ablation Module	Training rote	
		50%	80%
UC Merced	KD	76.07	93.62
	KD+TCSM	76.52(0.45↑)	94.17(0.55↑)
	KD+MCSM	78.14(2.07↑)	94.68(1.06↑)
	KD+TCSM+MCSM	78.79(2.72↑)	95.47(1.85↑)
Dataset	Ablation Module	Training rote	
		20%	50%
AID	KD	89.42	93.15
	KD+TCSM	90.25(0.83↑)	93.47(0.32↑)
	KD+MCSM	91.02(1.60↑)	93.95(0.80↑)
	KD+TCSM+MCSM	91.79(2.54↑)	94.44(1.29↑)

respectively. These ablation studies were conducted by incrementally incorporating the MCSM and TCSM modules into the basic distillation model. The table presents the distillation accuracy and the effects of improvement when the MCSM and TCSM modules are applied individually to a conventional knowledge distillation model. The results demonstrate that both the MCSM and TCSM modules significantly enhance distillation performance, and the effect is more significant when there is less training data. Compared with TCSM, the effect of the MCSM module is more pronounced. The low proportion of data with variable target sizes affects the degree to which TCSM promotes accuracy.

Next, we will conduct an ablation study to investigate the optimal setting of the parameters β_1 and β_2 in (2). We will observe changes in distillation accuracy by varying β_1 and β_2 to values of 0, 0.5, 1.0, and 1.5, respectively. For this experiment, the NWPU-RESISC45 dataset was selected for training, utilizing ResNet50 as the teacher network. MobileNetV2, GoogleNet, and SqueezeNet1_0 were chosen as the student networks.

From Fig. 6, it is evident that as β_1 increases from 0 to 1, the accuracy of model distillation improves significantly.

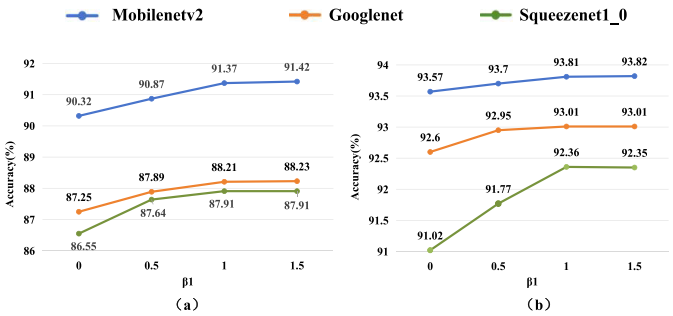


Fig. 6. Ablation analysis of the MCSM parameter factor β_1 based on the NWPU-RESISC45 dataset. (a) Variation in distillation accuracy when the training dataset proportion is 10% and (b) variation in distillation accuracy when the training dataset proportion is 20%.

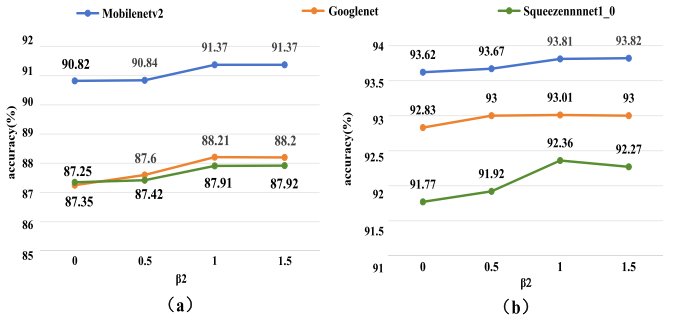


Fig. 7. Ablation analysis of the TCSM parameter factor β_2 based on the NWPU-RESISC45 dataset. (a) Variation in distillation accuracy when the training dataset proportion is 10% and (b) variation in distillation accuracy when the training dataset proportion is 20%.

However, when β_1 increases from 1 to 1.5, the improvement in model distillation accuracy becomes negligible, and a slight downward trend is observed in some experiments. Consequently, in this experiment, the parameter factor was set to 1. Fig. 7 illustrates the ablation study on the selection of β_2 values, with β_1 fixed at 1. Through the aforementioned ablation experiments, it can be concluded that the corresponding loss parameter β_2 in (2) is more appropriately set to 1.

V. EFFECTIVENESS ANALYSIS

To further validate the effectiveness of our proposed distillation method, we conducted a series of analyses to assess the performance of the distillation model based on the previously mentioned experiments. The effectiveness analyses were conducted using the UC Merced and AID datasets. Category activation maps and confusion matrices are selected as evaluation indicators.

A. Category Activation Maps

Fig. 8 displays five remote sensing scene images in the UC Merced and AID datasets, along with their category activation maps generated by students with different training modes. The scene categories, from left to right, are airport, bridge, church, sparse residential, and storage tanks. These images exhibit complex scene representations, featuring targets of varying scales and expansive scenes containing small targets.

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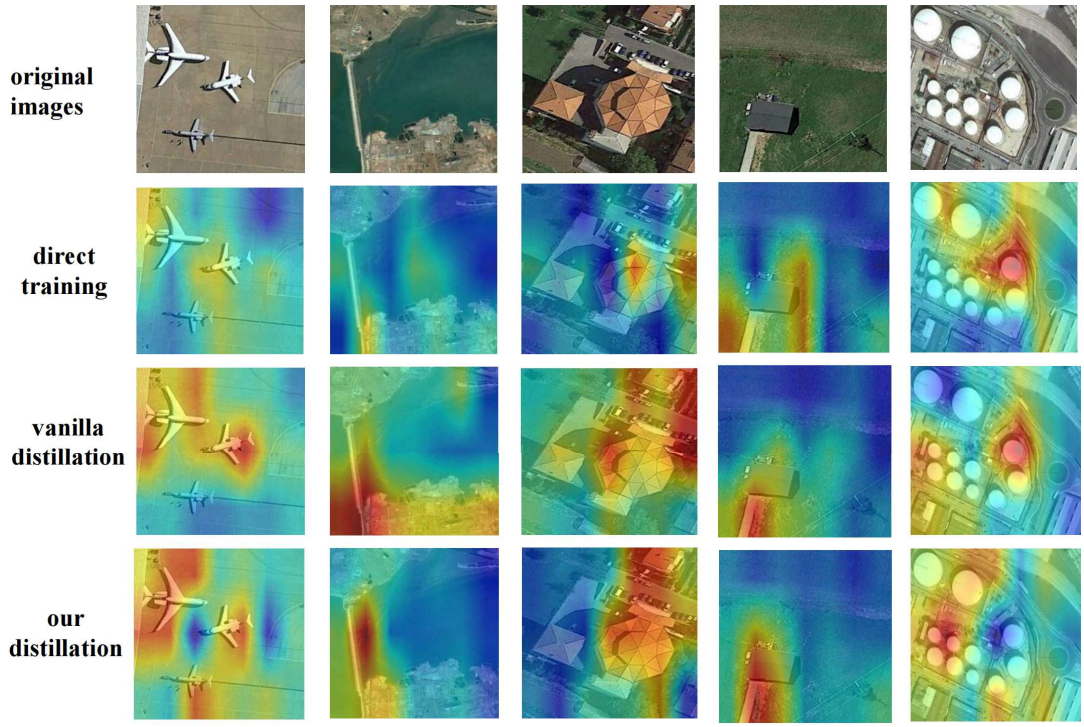


Fig. 8. Comparison of category activation maps with different training modes. The first row displays the original images, the second row presents the class activation maps generated by the student network by direct training, the third row illustrates the class activation maps of the student network obtained through vanilla distillation, and the fourth row showcases the class activation maps of the student network obtained by our proposed method.

The second row displays the category activation maps generated when the original remote sensing images are processed by the directly trained SqueezeNet1_0. The third row shows the category activation maps resulting from SqueezeNet1_0 trained by vanilla distillation. The fourth row illustrates the category activation maps obtained from SqueezeNet1_0 trained by our proposed distillation method.

From the figure, it is evident that the network model trained using our proposed distillation method captures significant features in complex scenes and is more effective than those trained through direct training or vanilla distillation methods. Specifically, when identifying scenes with targets of varying scales, such as airports and storage tanks, the distillation model proposed in this article enables student networks to recognize targets of different sizes more effectively than direct training and conventional distillation approaches. Furthermore, this model allows student networks to accurately locate discriminative targets and their surrounding areas in large scenes when identifying small targets, such as spark residential. In addition, in complex scenarios such as bridges and churches, the proposed distillation model encourages the student network to focus more on the target category itself, followed by the feature representation of the surrounding environment, compared to direct training and conventional distillation methods. This figure illustrates that the distillation method proposed in this article enables the student network to more accurately capture image regions with discriminative features.

B. Confusion Matrix

In addition to the category activation maps, we further validate the effectiveness of the distillation model proposed in this article by examining the confusion matrix. We selected MobileNetV2 and ResNet18 as the student network and compared the confusion matrices obtained from the validation set using both vanilla distillation training and our proposed distillation method on the AID and UC Merced datasets.

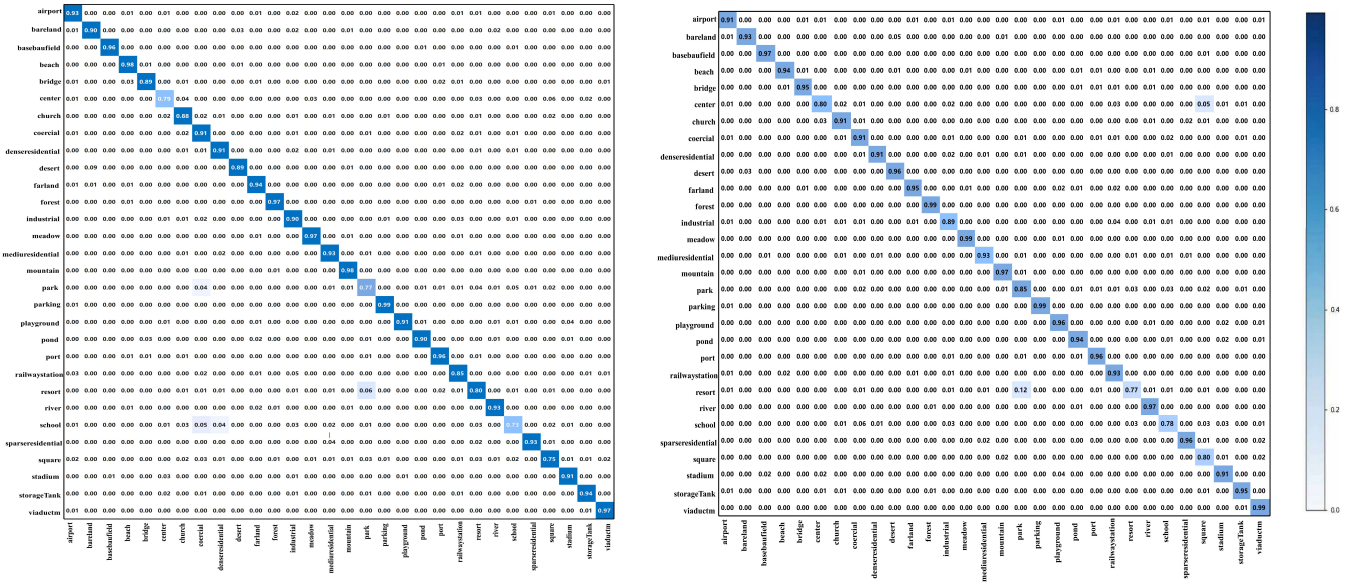
Fig. 9 illustrates the confusion matrices of Mobilenetv2 on the AID dataset. In this validation experiment, the ratio of the training set to the validation set was 2:8. Fig. 9(a) depicts the confusion matrix of MobileNetV2 trained through vanilla distillation on the validation set, and Fig. 9(b) presents the confusion matrix of MobileNetV2 trained with our proposed distillation model on the validation set.

By comparing the confusion matrices in Fig. 9, we observe that the MobileNetV2 network, when trained using the distillation method proposed in this article, exhibits a lower misclassification rate compared to those trained using vanilla distillation methods. This network demonstrates an enhanced ability to learn significant features of various categories within complex scenes. The performance improvement is particularly evident in the recognition of the following categories: “bridge,” “church,” “desert,” “park,” “pond,” “river,” “school,” and “square.”

Fig. 10 presents the confusion matrices for ResNet18 on the UC Merced dataset. In this validation experiment, the numerical ratio of the training set to the validation set is 8:2. Fig. 10(a) depicts the confusion matrix of ResNet18 trained using vanilla distillation on the validation set, while Fig. 10(b) illustrates the confusion matrix of ResNet18 trained using our proposed distillation method on the same validation set.

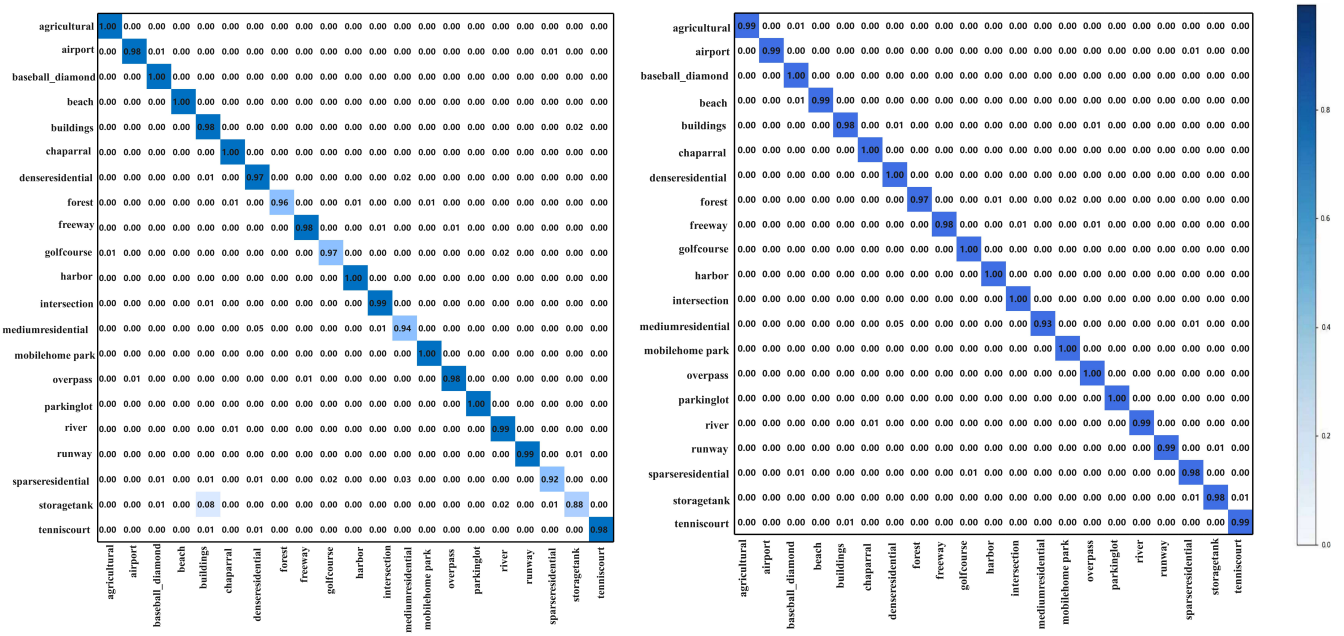
Similar to the conclusions drawn from Fig. 9, the confusion matrices presented in Fig. 10 indicate that the ResNet18

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(a) (b)

Fig. 9. Confusion matrices of MobileNetV2 on the AID dataset. (a) Confusion matrix of MobileNetV2 trained through vanilla distillation on the validation set and (b) confusion matrix of MobileNetV2 trained through our proposed distillation model on the validation set.



(a) (b)

Fig. 10. Confusion matrices of ResNet18 on the UC Merced dataset. (a) Confusion matrix of ResNet18 trained through vanilla distillation on the validation set and (b) confusion matrix of ResNet18 trained through our proposed distillation model on the validation set.

network, when trained using the distillation method proposed in this article, exhibits a lower misclassification rate compared to those trained using vanilla distillation methods. In addition, it demonstrates an enhanced ability to capture significant features of categories in complex scenes. This performance improvement is particularly pronounced in the recognition of the following categories: "sparse residential" and "storage tanks." These two categories are quintessential examples of remote sensing scene images characterized by large scenes with small targets and varying target sizes. This comparative experiment further substantiates the effectiveness of our proposed distillation model.

VI. Conclusion

To improve the performance of existing lightweight networks in remote sensing data interpretation tasks, a novel distillation model for remote sensing data classification is introduced. This model aims to transfer the generalization and cognitive abilities of the teacher network to the student network,

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which can be deployed on edge devices. The model incorporates two supervision modules: the MCSM and the TCSM. The MCSM emphasizes transferring the teacher network's understanding of interclass differences and intraclass similarities in remote sensing data to the student network. Simultaneously, the TCSM primarily enhances the student network's ability to recognize targets of varying sizes. The experimental results demonstrate that the distillation model significantly enhances the performance of student networks, even surpassing that of more complex network models with a larger number of parameters in terms of classification accuracy.

The proposed distillation model requires substantial computational resources. In future research, we will further explore the structured representation of data features. The structured representation of data will serve the knowledge distillation process, achieving high-performance distillation effects while reducing computational pressure, enabling more efficient distillation with fewer resources.

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